

LINEAR REGRESSION

SUBJECTIVE QUESTION

Submitted by :
Neine Arora

From your analysis of the categorical variables from the dataset, what could you infer about their effect on the dependent variable?

There were 7 categorical variables in dataset provided and Cnt or Count is the dependant variable.

Observations made are as below:

1. Count of total rental bikes increased from year 2018 to 2019
2. Count of total rental bikes usage is noted most in fall season followed by summer, winter and spring.
3. Count of total rental bikes usage is lower towards the start and end of months in year. It is highest in July month.
4. Count of total rental bikes usage is more when there is no holiday
5. Count of total rental bikes usage is more in 'Clear' weather
6. Count of total rental bikes usage does not vary much in weekdays or working days

Why is it important to use drop_first=True during dummy variable creation?

The number of dummy coded variables needed is one less than the number of possible values, which is $k-1$.

Drop_first when made True handles creation of dummy encoding variables for multiple categories. It follows the rule of $k-1$ where k is categorical levels. This helps in removing the first level.

Looking at the pair-plot among the numerical variables, which one has the highest correlation with the target variable?

Target Variable considered here is 'Cnt' or 'Count'

In pair plot, 'count' (Count of total rental bikes) and 'registered' (count of registered users) variables seems to have highest correlation. The scatter plot shows strong positive correlation. The heatmap executed as next step shows Pearson correlation coefficient (r value) as +0.95 between both variables.

How did you validate the assumptions of Linear Regression after building the model on the training set?

Once the linear regression model is determined and variables are finalised, we will use this model to predict data point for count (y : target variable).

Next step would be to determine the residuals by finding difference between original datapoints in count (y_train) variable and predicted data points of count variable (y_train_pred)

To validate the assumption of linear regression, these residuals should be normally distributed (Mean should be 0 and standard deviation should be 1) . We can see the same when we plot the distribution graph for residuals. The plot shows bell curve with centre at 0 (mean). Error terms also do not show any patterns and have constant variance.

Based on the final model, which are the top 3 features contributing significantly towards explaining the demand of the shared bikes?

The final model is as below:

$$\text{Count} = 0.273 + 0.235 * \text{Year} + 0.430 * \text{temp} - 0.149 * \text{windspeed} - 0.104 * \text{spring} + 0.039 * \text{winter} - 0.044 * \text{Jan} - 0.066 * \text{July} + 0.053 * \text{Sep} - 0.044 * \text{Sun} - 0.287 * \text{Light Snow} - 0.078 * \text{Mist} + \text{Cloudy}$$

Based on final model top three features contributing significantly towards explaining the demand are:

- Temperature (0.430)
- Light Snow (weathersit : Light Snow, Light Rain + Thunderstorm + Scattered clouds, Light Rain + Scattered clouds) (-0.287)
- Year (0.235)

So it recommended to give these variables utmost importance while planning to achieve maximum demand.

Explain the linear regression algorithm in detail

Below steps can be followed to perform linear regression:

Data Preparation (EDA and Variable Transformation)

After understanding the dataset, imputing missing values and dropping the columns not required, the rest of the variables need to be analysed further. Perform exploratory data analysis using univariate and bivariate analysis. Categorical variables need to go under One Hot encoding (binary data) and dummy variable encoding (multiple levels in variable) so that they can be analysed further.

Split the dataset – Train and Test

Split the data into two parts. First part will be used to create model and train model. The second part will be used to test the model accuracy later. The dataset should be divided into 70:30 ratio and

Rescaling the variables

Rescale the numerical variables for normalisation (Min Max Scaling) or Standardisation (standardising the variable by bringing mean = 0 and sigma = 1) using `scaler.fit_transform`

Model Building

Pop the target variable out of dataset and assign it to `y_train`. Use stepwise (front or backward) regression technique or RFE method to determine the variables of final model. In case of RFE, provide number of features to determine the model. Determine VIF in each iteration and eliminate the variables basis the high p value and high VIF.

Acceptable ranges for p values <0.05 and VIF <5 .

Residual Analysis

Create predicted values by using the final model. Determine the difference between train y and train_pred to derive residuals. Create distplot on these residuals to determine if residuals follows normal distribution. The indicator should be mean = 0 and bell curve showing normal distribution.

Making Predictions

Transform and scale test data using scaler.transform on numerical variables. Pop the y_test variable out of dataset and assign rest to X_test. Align the test dataset variables by dropping the variables not required and add constant to X_test dataset. Using the model finalised while training on this X_test dataset determine the y_pred.

Model Evaluation

Check the difference between R squared values of test set. Closer the R squared values of train and test set, better the model.

Regression Equations

Using the constant and coefficient of variables in final model, create an equation. This equation would suggest the relative importance of each predict (independent) variable in regards to target (dependent) variable.

Explain the Anscombe's quartet in detail.

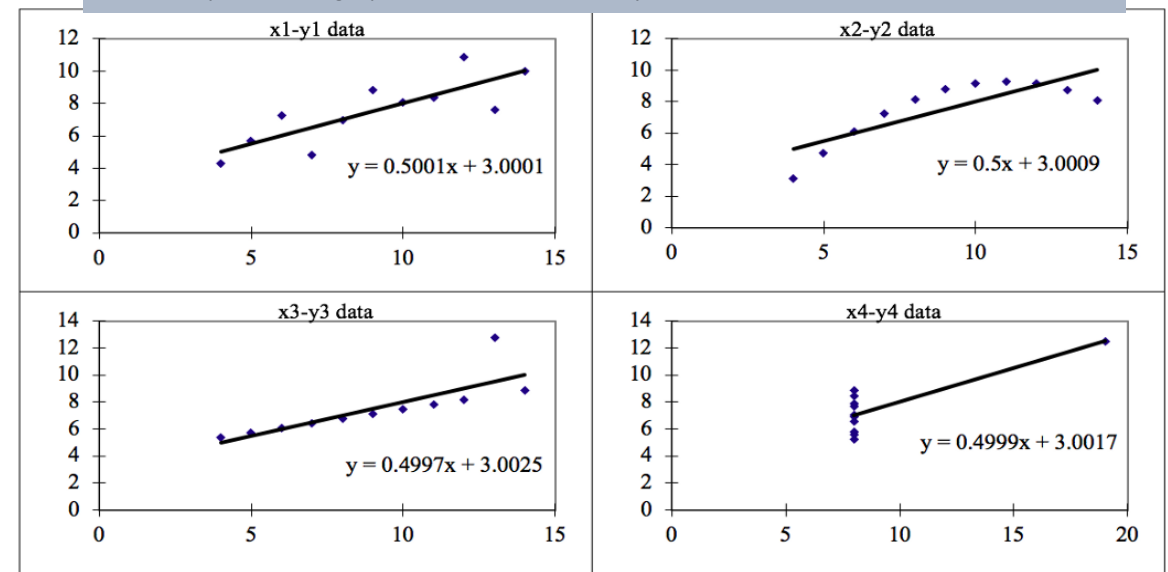
Anscombe's quartet comprises four data sets that have nearly identical simple descriptive statistics, yet have very different distributions and appear very different when graphed. They have very different distributions and appear differently when plotted on scatter plots.

It was constructed in 1973 by statistician Francis Anscombe to illustrate the importance of plotting the graphs before analyzing and model building, and the effect of other observations on statistical properties. There are these four data set plots which have nearly same statistical observations, which provides same statistical information that involves variance, and mean of all x,y points in all four datasets. Example is as follows

4 Different Datasets with similar summary statistics

Anscombe's Data											
Observation	x1	y1		x2	y2		x3	y3		x4	y4
1	10	8.04		10	9.14		10	7.46		8	6.58
2	8	6.95		8	8.14		8	6.77		8	5.76
3	13	7.58		13	8.74		13	12.74		8	7.71
4	9	8.81		9	8.77		9	7.11		8	8.84
5	11	8.33		11	9.26		11	7.81		8	8.47
6	14	9.96		14	8.1		14	8.84		8	7.04
7	6	7.24		6	6.13		6	6.08		8	5.25
8	4	4.26		4	3.1		4	5.39		19	12.5
9	12	10.84		12	9.13		12	8.15		8	5.56
10	7	4.82		7	7.26		7	6.42		8	7.91
11	5	5.68		5	4.74		5	5.73		8	6.89
				Summary Statistics							
N	11	11		11	11		11	11		11	11
mean	9.00	7.50		9.00	7.500909		9.00	7.50		9.00	7.50
SD	3.16	1.94		3.16	1.94		3.16	1.94		3.16	1.94
r	0.82			0.82			0.82			0.82	

Datasets plotted on graphs with same summary statistics shows different distribution

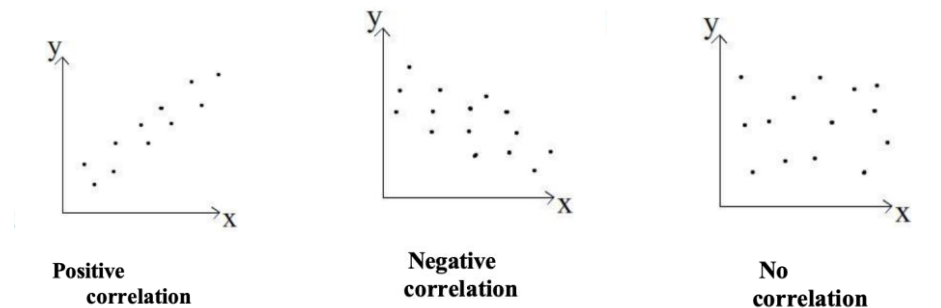


What is Pearson's R?

R is a correlation coefficient that measures the strength of the relationship between two variables, as well as the direction on a scatterplot. The value of r is always between a negative one and a positive one (-1 and a +1).

$$-1 \leq r \leq 1$$

This means that if $r = -1$, the data lies on a perfectly straight line with a negative slope. Also, if $r = +1$, the data lies on a perfectly straight line with a positive slope. If $r = 0$, this means there is no linear correlation.



r can be calculated as

$$r = \frac{n(\sum xy) - (\sum x)(\sum y)}{\sqrt{[n\sum x^2 - (\sum x)^2][n\sum y^2 - (\sum y)^2]}}$$

What is Pearson's R?

Usage: r is used to determine the strength of relationship between two continuous variables.

Strength of Association	Coefficient, r	
	Positive	Negative
Small	.1 to .3	-0.1 to -0.3
Medium	.3 to .5	-0.3 to -0.5
Large	.5 to 1.0	-0.5 to -1.0

Advantages:

- Shows the strength of two variables and whether there is indeed a relationship there
- Study behaviors of random independent variables
- It helps to gain quantitative data that's easy to analyse compared to more other complex statistical tests.

Disadvantages:

- It cannot show cause and effect
- It assumes that there is always some form of a linear relationship
- Correlation does not imply causation. Just because two independent variables have a high correlation does not mean that it is a causal relationship. (One variable causes the other to happen)

What is scaling? Why is scaling performed? What is the difference between normalized scaling and standardized scaling?

Feature scaling is used while data preparation or during pre-processing of data before creating a machine learning model.

Why performed: To be able to compare all numerical features in a similar scale, so that original features are not impacted due to difference in range or original range of each feature.

Techniques and differences:

The most common techniques of feature scaling are Normalization and Standardization.

Normalization (Min Max scaling) is used when we want to bound our values between two numbers, typically, between $[0,1]$ or $[-1,1]$.

While **Standardization** transforms the data to have zero mean and a variance of 1, they make our data **unitless**.

You might have observed that sometimes the value of VIF is infinite. Why does this happen?

VIF indicates the percentage of the variance inflated for each variable's coefficient. Beginning at a value of 1 (no collinearity), a VIF between 1–5 indicates moderate collinearity while values above 5 indicate high collinearity.

VIF shall be infinite only when R squared value is 1. This would mean that all variance has been explained through the model. **This might not be good or suggest high model accuracy due to below reason:**

The Coefficient of Determination (R-Squared value) explains how close the data is to the fitted regression line (the model). R-squared values range from 0 to 1. Generally, **a model with an R-Squared value closer to 1 has a lower MSE.** Mean Square Error (MSE) is a measure of how accurate predictions are by comparing the training data with the regression model to determine the average error of the predictions. MSE should be minimized to get a more accurate model. An MSE of 0 would mean that the model is **overfitting** the data, i.e. the model is too complex and will have poor performance.

Example: You may be predicting the temperature in Fahrenheit and one of your independent variables is temperature in Celsius. You would obviously get a very high R-Squared but a model that predicts one variable using the same variable in another form is useless.

What is a Q-Q plot? Explain the use and importance of a Q-Q plot in linear regression

Quantile-Quantile (Q-Q) plot, is a graphical tool to help us assess if a set of data plausibly came from some theoretical distribution such as a Normal, exponential or Uniform distribution. Also, it helps to determine if two data sets come from populations with a common distribution.

Use: A Q-Q plot helps you compare the sample distribution of the variable at hand against any other possible distributions graphically.

Importance in LR: This helps in a scenario of linear regression when we have training and test data set received separately and then we can confirm using Q-Q plot that both the data sets are from populations with same distributions.

Other examples: It is used to check following scenarios also:

If two data sets —

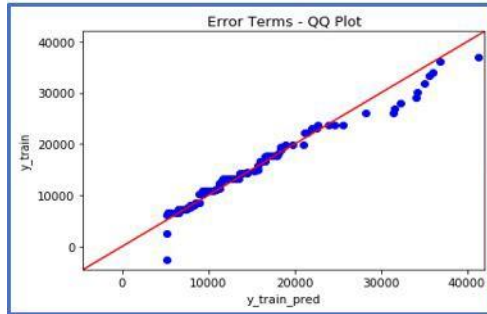
- I. come from populations with a common distribution
- II. have common location and scale
- III. have similar distributional shapes
- IV. have similar tail behavior

What is a Q-Q plot? Explain the use and importance of a Q-Q plot in linear regression

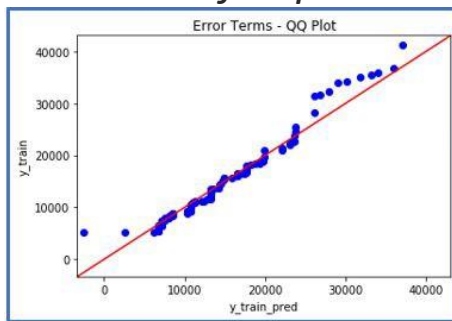
Below are the possible interpretations for two data sets.

*a) **Similar distribution:** If all point of quantiles lies on or close to straight line at an angle of 45 degree from x –axis*

*b) **Y-values < X-values:** If y-quantiles are lower than the x-quantiles.*



*c) **X-values < Y-values:** If x-quantiles are lower than the y-quantiles.*



*d) **Different distribution:** If all point of quantiles lies away from the straight line at an angle of 45 degree from x –axis.*